

RESEARCH AND EDUCATION

# Effect of finishing protocols on dentin surface characteristics and bond strength after tooth preparation for indirect restorations



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## ABSTRACT

**Statement of problem.** The impact of different finishing protocols following preparation for indirect restorations on surface roughness, smear layer, and bond strength remains unclear, with potential implications for clinical longevity.

**Purpose.** The purpose of this in vitro study was to evaluate the effect of different dentin finishing protocols on surface (roughness and morphology), smear layer thickness, and bond strength to a resin-matrix glass-ceramic.

**Material and methods.** Unused preparation instruments were characterized under scanning electron microscopy (SEM), and 144 human third molars were prepared and divided into groups: control (medium grit diamond rotary instruments, no finishing [CT]), fine/superfine diamond rotary instruments [FB], fine/superfine tungsten carbide burs [CB], fine/superfine ultrasonic diamond tips [UT], SiC/Al<sub>2</sub>O<sub>3</sub> mounted stones [MS], fine/superfine sintered diamond rotary instruments [SB], Al<sub>2</sub>O<sub>3</sub> airborne-particle abrasion [AO], and resin coating [RC]. Surface roughness and morphology were analyzed using laser scanning confocal microscopy and SEM. Smear layer thickness was assessed using SEM combined with the ImageJ software program, and bond strength was evaluated using microtensile testing. One-way analysis of variance (ANOVA) and post hoc tests were used for statistical analysis, followed by Pearson correlation. Failure modes were analyzed using the chi-squared test ( $\alpha=.05$ ).

**Results.** Significant differences in surface roughness, smear layer thickness, and bond strength were observed across the groups ( $P<.001$ ). All finishing protocols resulted in lower surface roughness and higher bond strength compared with the control. RC achieved the highest bond strength, followed by MS. A very weak correlation was found between surface roughness, smear layer thickness, and bond strength.

**Conclusions.** Finishing protocols affect dentin roughness, smear layer, and bond strength after tooth preparation. Resin coating and fine and superfine mounted SiC and Al<sub>2</sub>O<sub>3</sub> stones improved adhesion and may enhance outcomes for indirect restorations. (J Prosthet Dent 2026;135:371.e1-e11)

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## Clinical Implications

Optimizing finishing protocols is essential for improving the bond strength of indirect restorations. Supplementary finishing procedures performed after tooth preparation have demonstrated the potential to enhance dentin characteristics, which may contribute to the long-term durability of indirect restorations.

Indirect procedures are often required for restoring teeth with extensive structural damage or those needing modifications in form and function.<sup>1</sup> This process typically involves preparation with rotary instruments to create adequate space and draw for the restoration.<sup>2</sup> Various finishing techniques have been described to achieve smoother surfaces and minimize the smear layer thickness.<sup>1–7</sup> Although fine and superfine diamond rotary instruments have been commonly used, their irregular morphology may produce uneven surfaces.<sup>2,8,9</sup>

Alternative instruments such as diamond ultrasonic tips, multibladed tungsten carbide burs, mounted stones, and aluminum oxide ( $\text{Al}_2\text{O}_3$ ) have also been suggested.<sup>2,9–15</sup> These methods produce varying effects on roughness and bond strength.<sup>4,6,10,14–16</sup> Resin coating (or immediate dentin sealing) has been recommended after preparation to protect dentin and preserve bond strength before interim restoration placement.<sup>17–23</sup> This straightforward technique may improve bonding for both direct and indirect restorations.<sup>19</sup> However, its effectiveness in single-session indirect restorations, such as chairside computer-aided design and computer-aided manufacturing (CAD-CAM) procedures, remains underexplored.<sup>10,23</sup>

Despite the lack of a clear correlation between roughness and bond strength,<sup>8</sup> the effects of different finishing instruments after preparations remain clinically relevant. With various finishing methods available, questions remain about their effects on surface characteristics, smear layer formation, and bond strength of indirect restorations to dentin.<sup>24</sup>

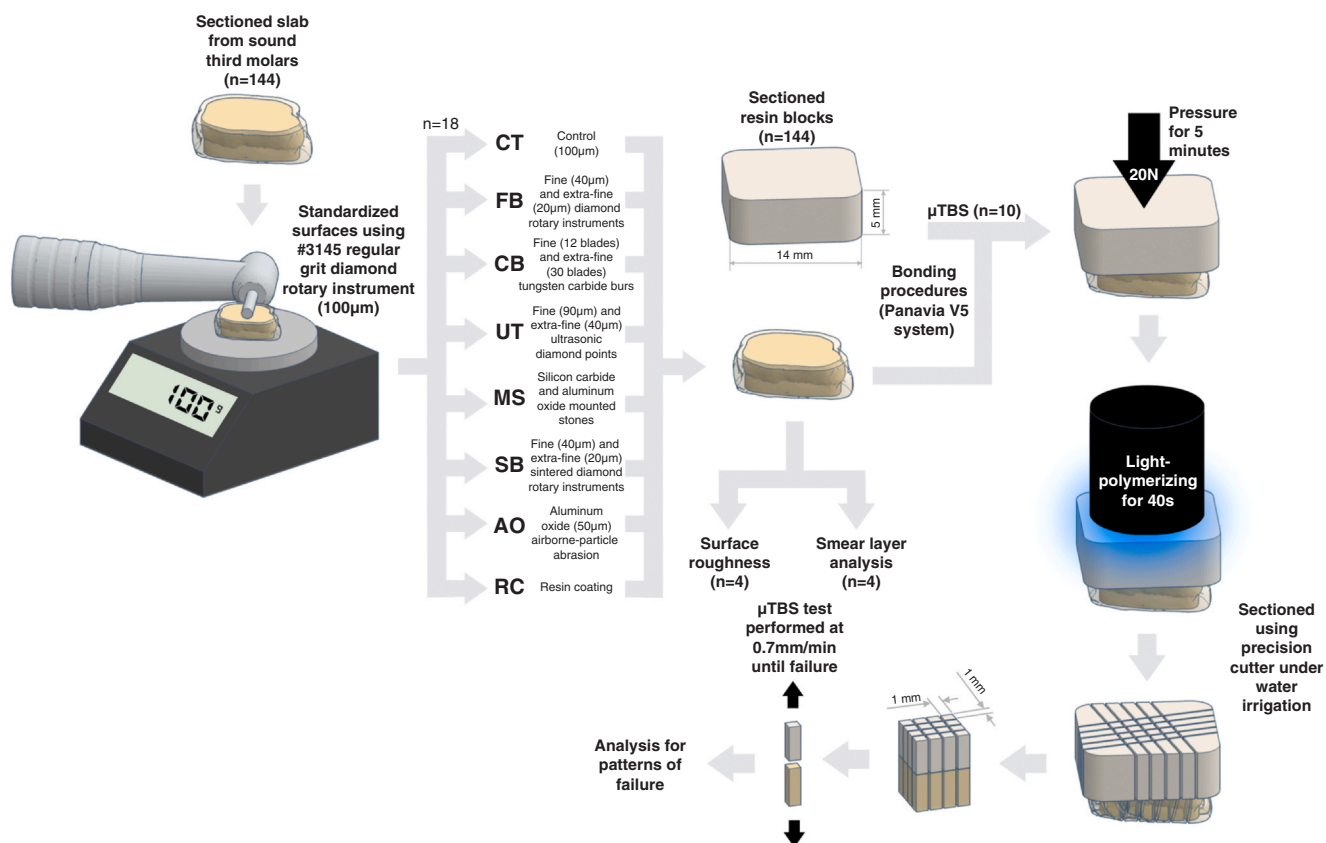
Therefore, the purpose of this study was to evaluate the effect of different dentin finishing protocols after indirect preparations on surface characteristics (roughness and morphology), smear layer thickness, and bond strength to a resin-matrix glass-ceramic restoration. The research hypotheses were that different finishing protocols would influence all tested parameters, that a reduction in dentin surface roughness and smear layer thickness would increase bond strength, and that resin coating would enhance bond strength to dentin.

## MATERIAL AND METHODS

Unused preparation instruments ( $n=1$ ) were sonicated in distilled water for 15 minutes, dried, mounted on aluminum stubs, and examined under scanning electron microscopy (SEM) (VEGA 3 LMU; TESCAN) at  $\times 100$  and  $\times 300$  magnifications. Sample size was determined through power analysis based on pilot experiments and literature data.<sup>2,25,26</sup> For bond strength, each group included 10 specimens, each yielding approximately 16 bonded sticks. Based on large effect size (Cohen  $f$  approximately=16.92), power  $>.999$  was achieved at  $\alpha=.05$ . For surface roughness and smear layer thickness, 4 specimens per group were used, which was consistent with similar studies.<sup>2,8,24</sup> These also showed high effect (Cohen  $f=4.21$  and  $14.27$ , respectively), ensuring power  $>.999$ .

A total of 144 sound freshly extracted human third molars were selected after ethics approval (Research Ethics Committee No. 4 938 277). The teeth were cleaned and stored following the International Organization for Standardization (ISO) 11405 standard.<sup>27</sup> Dentin surfaces were exposed by removing the occlusal third of the crowns (2 mm) using a precision saw (Isomet 1000; Buehler) and the roots embedded in polystyrene resin (AM 190; AeroJet) for handling. The surfaces were flattened using SiC papers (#600), followed by simulated preparation using medium-grit cylindrical diamond rotary instrument (#3145; KG Sorensen) in a high-speed electric handpiece (1:5, WG-99 LT + EM 12-L; W&H) at 10 000 RPM and 0.6-Nm torque. The handpiece was secured to a device ensuring standardized instrument contact (0.98 to 1.03 N) with the specimen surface (Fig. 1). The specimens were randomly assigned to finishing protocol groups ( $n=18$ ) using a computer-generated randomization list (Excel 365; Microsoft Corp): control (no finishing [CT]); fine and superfine diamond rotary instruments [FB]; fine (12 blades) and superfine (30 blades) tungsten carbide burs [CB]; fine and superfine ultrasonic diamond tips [UT]; fine and superfine mounted SiC and  $\text{Al}_2\text{O}_3$  stones [MS]; fine and superfine sintered diamond rotary instruments [SB];  $\text{Al}_2\text{O}_3$  airborne-particle abrasion (50  $\mu\text{m}$ ) [AO]; and resin coating [RC]. All groups underwent the same standardized initial dentin preparation using SiC abrasive papers, followed by surface reduction with medium-grit diamond rotary instruments (Table 1).

For surface roughness, specimens ( $n=4$ ) underwent ultrasonic cleaning in distilled water for 15 minutes, followed by gentle air drying. Surface roughness was analyzed immediately using a laser scanning confocal microscope (LEXT 4000; Olympus Co) at  $\times 10$  magnification. Measurements were made at 4 equidistant points per specimen, and the average  $S_a$  ( $\text{mm}^2$ ) values were



**Figure 1.** Schematic diagram of study design. Device standardized for specimen preparation, controlling contact force (0.98 to 1.03 N) and instrument position. Dentin finishing protocols evaluated for surface roughness and smear layer thickness. Bonding procedures followed by microtensile bond strength testing ( $\mu$ TBS) and failure mode analysis.

calculated per the ISO 4287:2002 standard.<sup>28</sup> For SEM characterization, specimens were dehydrated in graded ethanol (50%, 70%, and 95% for 10 minutes each, followed by 30 minutes in 100% ethanol), then stored for 48 hours in a vacuum-sealed chamber with silica gel. Afterward, they were mounted on stubs, gold-sputtered, and imaged at  $\times 200$  and  $\times 1000$  magnifications (VEGA 3 LMU; TESCAN).

To evaluate smear layer thickness, additional specimens ( $n=4$ ) were cross-sectioned perpendicularly to the prepared surface, cleaned with air-water spray, and prepared for SEM as described. Micrographs were obtained at  $\times 5000$  magnification (VEGA 3 LMU; TESCAN). Smear layer thickness was measured at 10 equidistant points along each cross-section using an image-processing software program (ImageJ, v. 1.8.0; National Institutes of Health), and average values were calculated per group and quantitatively analyzed.

For the bond strength test, resin-matrix glass-ceramic CAD-CAM blocks (Brava Block; FGM) were sectioned into  $14 \times 14 \times 5$  mm pieces and airborne-particle abraded with  $50\text{-}\mu\text{m}$   $\text{Al}_2\text{O}_3$  particles at 0.5 MPa, ultrasonically cleaned for 3 minutes, coated with 10-methacryloyloxydecyl dihydrogen phosphate (MDP) silane

(Clearfil Ceramic Primer Plus; Kuraray), and air-dried for 30 seconds. The teeth were prepared ( $n=10$ ), cleaned with air-water spray, and dried. A self-etch adhesive (Tooth Primer, Panavia V5; Kuraray Noritake Dental) was applied for 20 seconds and gently air-dried. A dual-polymerizing resin luting agent (Panavia V5; Kuraray Noritake Dental) was dispensed via Automix tips, applied to restorations, placed on tooth surfaces, and held under a 20-N load for 5 minutes. Excess luting agent was removed, followed by 40-second photoactivation per surface (VALO Grand; Ultradent Products, Inc) (Fig. 1, Table 2).

The specimens were stored in distilled water at  $37^\circ\text{C}$  for 24 hours then sectioned perpendicularly along the X and Y axes under copious water irrigation (Fig. 1). Each sectioned tooth yielded approximately 16 rectangular stick-shaped specimens, each with a cross-sectional area of approximately  $1\text{ mm}^2$  at the adhesive interface (8 to 10 mm length), using only central tooth regions to minimize variability. The specimens were fixed to a Geraldelli jig (OG08; Odeme Dental Research) using cyanoacrylate cement (Super Bonder; Loctite) and tested in a microtensile tester (OM100; Odeme Dental Research), under 0.7 mm/minute stress until failure (Fig. 1).

**Table 1.** Dentin finishing protocols used after tooth preparation (steps described for each experimental group performed after initial preparations as in CT group)

| Group | Protocol                                                   | Characteristics                                                                                                                                                                   | Manufacturer            | Technique                                                                                                                                                                                                                                 |
|-------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CT    | Control;<br>Medium-grit diamond rotary<br>instruments      | #3145 (100 µm)                                                                                                                                                                    | KG                      | Contra angle (1:5) mounted on electric micromotor at 10 000 RPM; coupled to device to control contact force, 60 seconds each instrument under abundant irrigation.                                                                        |
| FB    | Fine and extra-fine diamond<br>rotary instruments          | #3145 F (40 µm);<br>#3145 FF (20 µm)                                                                                                                                              | Sorensen                | Contra angle (1:5) mounted on electric micromotor at 10 000 RPM; coupled to device to control contact force, 30 seconds each instrument under abundant irrigation.                                                                        |
| CB    | Fine and extra-fine tungsten<br>carbide burs               | 283 Carbide FG (12 blades);<br>9572FF Carbide FG (30 blades)                                                                                                                      | Prima Dental, Ângelus   | Contra angle (1:5) mounted on electric micromotor at 10 000 RPM; coupled to device to control contact force, 30 seconds each instrument under abundant irrigation.                                                                        |
| UT    | Fine and extra-fine ultrasonic<br>diamond tips             | CR4 (90 µm);<br>CR4U (40 µm)                                                                                                                                                      | CVDentus                | Piezoelectric ultrasound (DentSurg PRO; CVDentus); 20 mL/minute water flow rate; 80% maximum power for CR4; 50% maximum power for CR4U; coupled to device to control contact force, 30 seconds each instrument under abundant irrigation. |
| MS    | Silicon carbide and aluminum<br>oxide mounted stones       | Durastone Green (Silicon Carbide with<br>ceramic binder-SiC) (PW1 504P);<br>Durastone White (Aluminum Oxide with<br>ceramic binder-Al <sub>2</sub> O <sub>3</sub> ) (FG199-016-W) | Dhpro                   | Contra angle (1:5) mounted on electric micromotor at 10 000 RPM; coupled to device to control contact force, 30 seconds each instrument under abundant irrigation.                                                                        |
| SB    | Fine and extra-fine sintered<br>diamond rotary instruments | FG199-016-A (40 µm);<br>FG199-016-C (20 µm)                                                                                                                                       | Dhpro                   | Tip of device perpendicularly positioned 10-mm from surface with 0.5 MPa for 30 seconds                                                                                                                                                   |
| AO    | Aluminum oxide airborne-particle<br>abrasion               | Al <sub>2</sub> O <sub>3</sub> - 50 µm                                                                                                                                            | BioArt                  | Clearfil SE: Active Primer application for 20 seconds, light air stream for 30 seconds from 100 mm, bond application with even film, light air stream.                                                                                    |
| RC    | Resin coating                                              | Clearfil SE Bond 2 + Clearfil AP-X Esthetic A2D<br>flowable composite resin                                                                                                       | Kuraray Noritake Dental | Clearfil AP-X Esthetic: Thin layer applied with disposable brush and light-activated for 40 seconds (VALO Grand; Ultradent Products, Inc).<br>*No additional finishing procedures were performed after the resin coating.                 |

Maximum load (N) at failure was recorded, and bond strength calculated based on the adhesive area (N/mm<sup>2</sup>, MPa).<sup>26</sup>

Fractured specimens were examined under an optical microscope (MS5; Leica Microscopy Systems Ltd) at ×10 magnification and classified as adhesive in dentin, adhesive in resin, adhesive mixed, cohesive in dentin, cohesive in resin, or mixed.<sup>25</sup> Representative specimens were analyzed under SEM at ×200 magnification (VEGA 3 LMU; TESCAN).

Data distributions for surface roughness, smear layer thickness, and bond strength were assessed using the Shapiro-Wilk and Levene tests. The assumptions for 1-way analysis of variance (ANOVA) were verified and met for all tested variables. One-way ANOVA was performed, with the Dunnett test used to compare experimental groups with the control and the Tukey HSD test for multiple group comparisons. Pearson correlation evaluated relationships between surface roughness, smear layer thickness, and bond strength. Failure mode distribution was analyzed using the chi-squared test (α=.05). All statistical analyses were performed using a statistical software suite (GraphPad Prism, V. 10.2.3 (347); GraphPad Software LLC). SEM images for surface morphology were analyzed qualitatively.

**RESULTS**

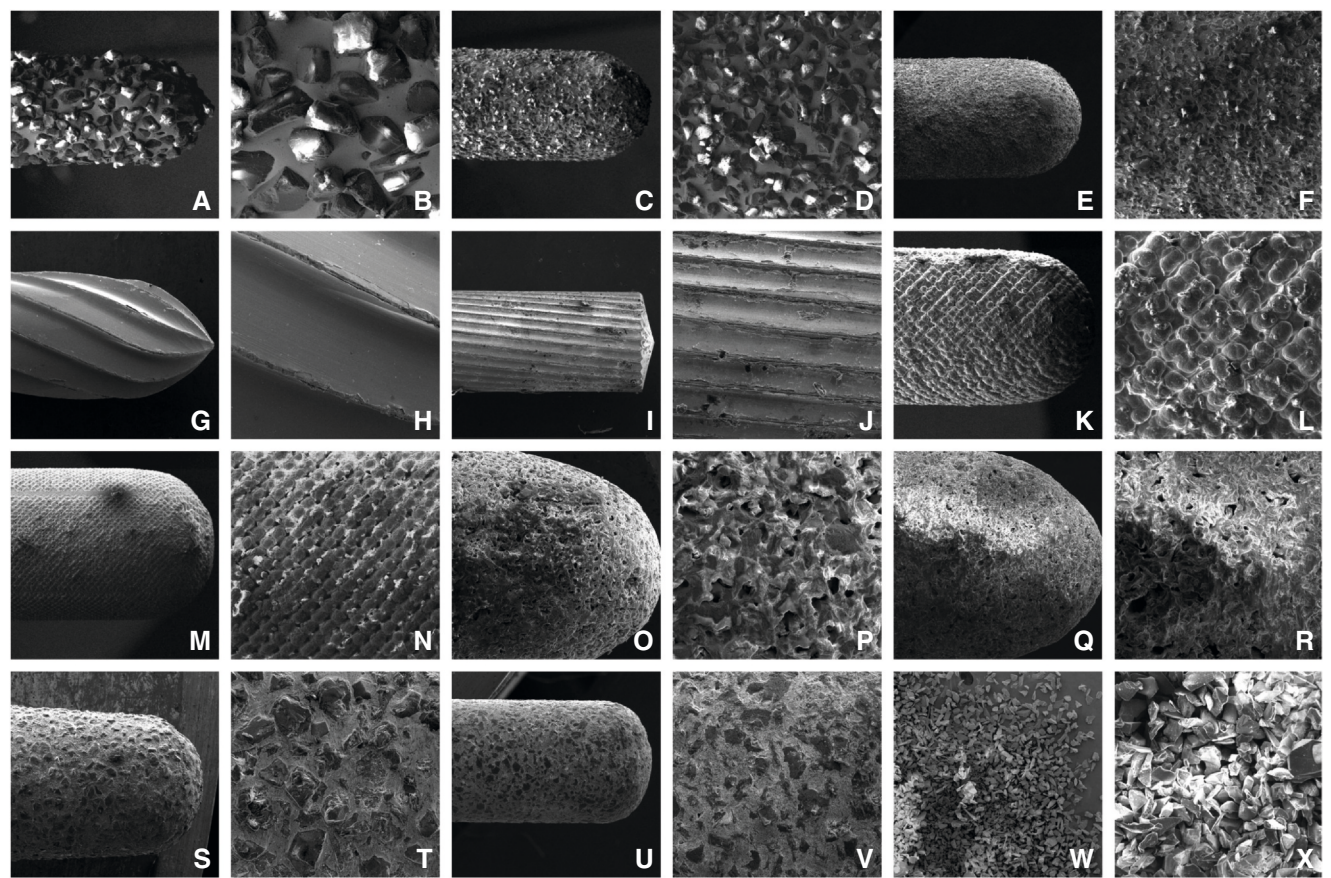
The comparative analysis of the instruments revealed distinct morphological characteristics. [Figure 2](#) presents images at ×100 and ×300 magnifications of unused instruments in different regions. The CT group exhibited large, irregular diamond particles heterogeneously distributed ([Fig. 2A, B](#)). The FB group showed smaller, also irregular particles aggregated on the exposed metal rod ([Fig. 2C-F](#)). The CB group instruments exhibited distinct characteristics: one had an ogival-shaped active tip with homogeneously allocated transverse blades ([Fig. 2G, H](#)), while the other featured a 90-degree angled tip with a greater number of fine transverse blades, also homogeneously allocated. Slight cracks were observed on the edges of the 90-degree angle, along with pores on the surface of the second instrument ([Fig. 2I, J](#)). The UT group displayed a rounded active tip with a uniform diamond coating. Large, rounded diamond particles formed a well-organized and evenly distributed crystalline structure, completely coalescent with no uncovered areas ([Fig. 2K-N](#)). The MS group instruments had rounded active tips composed of SiC ([Fig. 2O, P](#)) or Al<sub>2</sub>O<sub>3</sub> ([Fig. 2Q, R](#)) particles, agglutinated by a ceramic binder on a metal rod. Discrete microporosities were present on both instruments, with granules embedded within the binder along with slight surface irregularities and deep valleys. In the SB group, the first instrument

**Table 2.** Materials used

| Material                                                                      | Manufacturer                       | Lot#                 | Composition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------|------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resin matrix glass-ceramic:<br>BRAVA Block, A1-LT/ 14 L<br>Panavia V5 - White | FGM<br><br>Kuraray Noritake Dental | 170522<br><br>960014 | Methacrylate monomers, initiator, co-initiator, stabilizers, silane, glass-ceramic particles, silica, and pigments.<br>Paste A – Bisphenol A glycidyl methacrylate (Bis-GMA), triethyleneglycol dimethacrylate (TEGDMA), hydrophilic aromatic dimethacrylate, hydrophilic aliphatic dimethacrylate, initiators, accelerators, silanated barium glass filler, silanated fluoroaluminosilicate glass filler, colloidal silica.<br>Paste B – Bisphenol A glycidyl methacrylate (Bis-GMA), hydrophilic aromatic dimethacrylate, hydrophilic aliphatic dimethacrylate, silanated barium glass filler, silanated aluminum oxide filler, accelerators, dl-camphorquinone, pigments.<br>10-methacryloyloxydecyl dihydrogen phosphate (MDP), 2-hydroxyethyl methacrylate (HEMA), hydrophilic aliphatic dimethacrylate, accelerators, water.<br>3-methacryloxypropyl trimethoxysilane, 10-methacryloyloxydecyl dihydrogen phosphate (MDP), ethanol. |
| Tooth Primer                                                                  | Kuraray Noritake Dental            | 9T0100               | Silanated barium glass filler, silanated silica filler, triethyleneglycol dimethacrylate (TEGDMA), hydrophosphoric aromatic dimethacrylate, dl-camphorquinone, initiators, accelerators.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Clearfil ceramic primer plus                                                  | Kuraray Noritake Dental            | 4J0078               | Primer – 10-methacryloyloxydecyl dihydrogen phosphate (MDP), 2-hydroxyethyl methacrylate (HEMA), hydrophilic aliphatic dimethacrylate, dl-camphorquinone, water.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Clearfil AP-X Esthetics Flow – A2D                                            | Kuraray Noritake Dental            | 5N0015               | Bond – 10-methacryloyloxydecyl dihydrogen phosphate (MDP), bisphenol A glycidyl methacrylate (Bis-GMA), 2-hydroxyethyl methacrylate (HEMA), hydrophobic aliphatic dimethacrylate, dl-camphorquinone, initiators, accelerators, silanated colloidal silica.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Clearfil SE bond                                                              | Kuraray Noritake Dental            | 210398               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

had a conical shape (Fig. 2S, T), while the second exhibited a cylindrical contour (Fig. 2U, V). Both had rounded active tips; the first instrument featured

irregularly shaped diamond particles homogeneously distributed across the surface, while the second had smaller diamond crystals with similar morphology. No



**Figure 2.** Scanning electron micrographs of instruments used (original magnifications  $\times 100$  and  $\times 300$ ). A, B, CT (medium-grit diamond rotary instrument). C, D, FB (fine-grit diamond rotary instruments). E, F, FB (superfine diamond rotary instruments). G, H, CB (fine tungsten carbide bur). I, J, CB (superfine tungsten carbide bur). K, L, UT (fine ultrasonic diamond tip). M, N, UT (superfine ultrasonic diamond tip). O, P, MS (silicon carbide with ceramic binder). Q, R, MS (aluminum oxide with ceramic binder). S, T, SB (fine sintered diamond rotary instruments). U, V, SB (superfine sintered diamond rotary instruments). W, X, AO (50  $\mu\text{m}$   $\text{Al}_2\text{O}_3$  particles).

visible regions of metal rod exposure were detected. The AO group exhibited  $\text{Al}_2\text{O}_3$  particles with significant size variations, ranging up to 50  $\mu\text{m}$ , with a small number exceeding this size and displaying pointed shapes (Fig. 2W, X).

The mean surface roughness values are presented in Table 3. One-way ANOVA revealed significant differences across the groups ( $P<.001$ ). The highest surface roughness values were observed for the CT group, which did not differ statistically from RC group ( $P=.223$ ). The AO group showed no statistical difference from the RC ( $P=.973$ ) and FB groups ( $P=.882$ ). The lowest roughness values were found for CB, UT, MS, and SB groups, which did not differ statistically from the FB group ( $P>.05$ ). The Dunnett post hoc test confirmed that only the RC group showed no significant difference compared to the CT (control) group ( $P=.087$ ).

SEM revealed distinct morphological characteristics of the dentin surfaces depending on the finishing protocols used (Fig. 3). The CT group exhibited pronounced surface irregularities, with elevated peaks, deep depressions, and non-uniform grooves, confirming its high surface roughness (Fig. 3A, B). The FB group had lower peaks and depressions than CT (Fig. 3C, D). The UT group showed evenly distributed irregularities, with deep and rounded peaks and valleys (Fig. 3G, H). The AO group led to significant deposition of aluminum oxide particles on the surface (Fig. 3M, N). In the RC group, dentin was covered by a resin-based material containing inorganic particles and porosities (Fig. 3O, P), compatible with the observed high surface roughness. The CB (Fig. 3E, F), MS (Fig. 3I, J), and SB (Fig. 3K, L) groups resulted in smoother, more homogeneous surfaces with small irregularities. Findings from SEM were consistent with confocal laser microscopy, providing complementary insights (Fig. 4). Notably, CB group confocal images revealed specific regions with deep valleys, where SEM at  $\times 5000$  magnification (Fig. 5) further identified microfissures in peritubular dentin.

The mean smear layer thickness values ( $\mu\text{m}$ ) are presented in Table 3. The 1-way ANOVA revealed significant differences across the groups ( $P<.001$ ). Based on the Tukey HSD test, the FB group exhibited the thickest

smear layer, with no statistical difference from the AO ( $P=.315$ ) and RC ( $P=.383$ ) groups. The CT group followed, showing no statistical difference from AO ( $P=.594$ ) and RC ( $P=.515$ ). The subsequent order was CB, UT, MS, and SB, with MS showing no statistical difference from UT ( $P=.109$ ) and SB ( $P=.942$ ). According to the Dunnett test, only AO ( $P=.298$ ) and RC ( $P=.240$ ) groups showed no significant difference compared to CT (control) group. Differences in smear layer thickness formation were also evident in the qualitative comparison (Fig. 6). Although the CB group exhibited a thinner smear layer, surface microcracks were observed in dentin, which became clearer in cross-section assessments (Fig. 5A, B). Pearson correlation analysis suggested a strong correlation between surface roughness and smear layer thickness ( $R^2=0.665$ ).

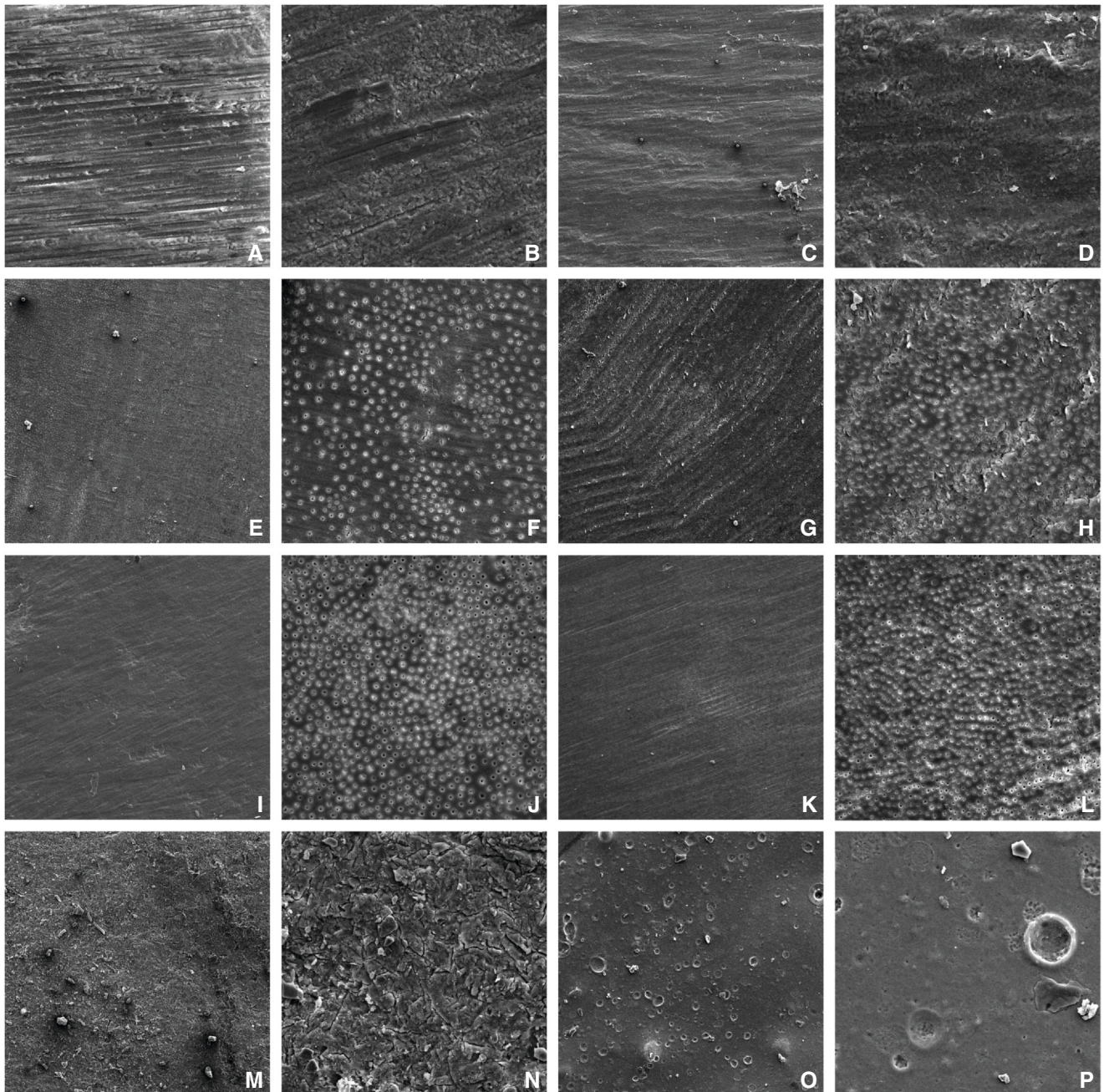
The mean bond strength values are shown in Table 3. One-way ANOVA revealed significant differences across the dentin finishing protocols ( $P<.001$ ). The Tukey HSD test indicated the highest bond strength values for RC, followed by MS, AO, FB, UT, and CB, in that order. The CT group exhibited the lowest bond strength values ( $P<.001$ ). The FB group showed no statistical difference from AO ( $P=.387$ ) or UT ( $P=.494$ ), while UT did not differ statistically from CB ( $P=.120$ ). The Dunnett post hoc test found no similarity between any experimental group and the CT (control) group ( $P<.001$ ). Pearson correlation analysis revealed a very weak correlation between surface roughness and bond strength ( $R^2=0.004$ ), as well as smear layer thickness and bond strength ( $R^2=0.001$ ).

The failure mode distribution for the groups is summarized in Figure 7. The chi-squared test revealed significant associations in the distribution of failure modes based on the different finishing protocols. Adhesive failures in dentin were most frequent for the CT (54%), FB (56%), CB (57%), and UT (51%) groups, with some cases showing dentin completely free of resin cement. Mixed adhesive failures were more prevalent for the AO (57%) and SB (47%) groups, while adhesive failures in resin were dominant for RC (59%) and MS (41%) groups. CB group was the only group with no cohesive failures.

**Table 3.** Mean surface roughness ( $\text{Sa} - \text{mm}^2$ ), smear layer thickness ( $\mu\text{m}$ ), bond strength (MPa) values, and standard deviation ( $\pm$ ) according to dentin finishing protocols

| Groups | Surface roughness ( $\text{mm}^2$ ) | Smear layer thickness ( $\mu\text{m}$ ) | Bond strength (MPa)            |
|--------|-------------------------------------|-----------------------------------------|--------------------------------|
| CT     | 1.52 $\pm$ 0.09 <sup>A</sup>        | 7.01 $\pm$ 0.32 <sup>B</sup>            | 10.79 $\pm$ 0.54 <sup>F</sup>  |
| FB     | 1.01 $\pm$ 0.05 <sup>BC</sup>       | 7.63 $\pm$ 0.28 <sup>A</sup>            | 18.12 $\pm$ 1.07 <sup>CD</sup> |
| CB     | 0.66 $\pm$ 0.06 <sup>C</sup>        | 4.27 $\pm$ 0.27 <sup>C</sup>            | 15.33 $\pm$ 0.63 <sup>E</sup>  |
| UT     | 0.74 $\pm$ 0.06 <sup>C</sup>        | 2.89 $\pm$ 0.12 <sup>D</sup>            | 16.95 $\pm$ 1.55 <sup>DE</sup> |
| MS     | 0.69 $\pm$ 0.09 <sup>C</sup>        | 2.47 $\pm$ 0.27 <sup>DE</sup>           | 27.97 $\pm$ 1.71 <sup>B</sup>  |
| SB     | 0.73 $\pm$ 0.15 <sup>C</sup>        | 2.30 $\pm$ 0.29 <sup>E</sup>            | 16.06 $\pm$ 1.37 <sup>E</sup>  |
| AO     | 1.15 $\pm$ 0.05 <sup>B</sup>        | 7.29 $\pm$ 0.52 <sup>*AB</sup>          | 19.39 $\pm$ 1.22 <sup>C</sup>  |
| RC     | 1.25 $\pm$ 0.36 <sup>*AB</sup>      | 7.30 $\pm$ 0.45 <sup>*AB</sup>          | 32.10 $\pm$ 1.79 <sup>A</sup>  |

\* Indicates statistical similarity between Control (CT) and experimental groups; Dunnett test ( $P>.05$ ). Distinct uppercase letters indicate statistical differences across groups in vertical; Tukey HSD test ( $P<.05$ ).



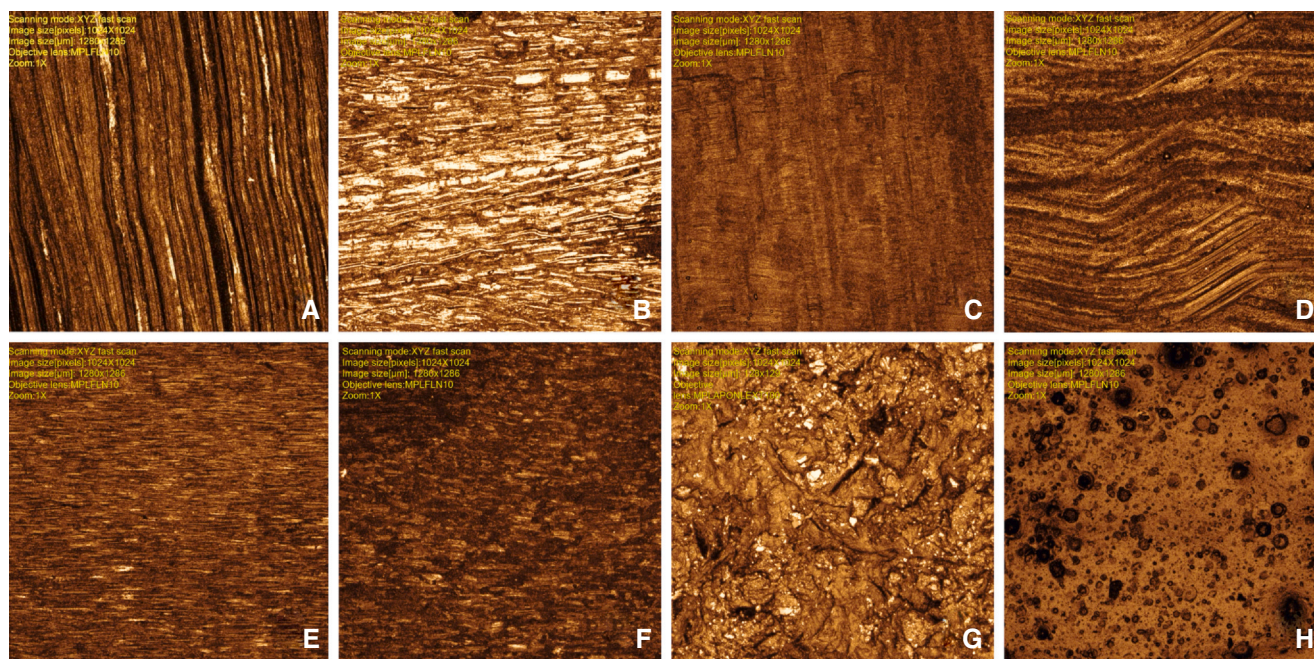
**Figure 3.** Scanning electron micrographs of dentin surfaces after finishing protocols (original magnification  $\times 200$  and  $\times 1000$ ). A, B, CT (medium-grit diamond rotary instruments). C, D, FB (fine and superfine diamond rotary instruments). E, F, CB (fine and superfine tungsten carbide burs). G, H, UT (fine and superfine ultrasonic diamond tips). I, J, MS (silicon carbide and aluminum oxide with ceramic binders). K, L, SB (fine and superfine sintered diamond rotary instruments). M, N, AO ( $50\ \mu\text{m}\ \text{Al}_2\text{O}_3$  particles). O, P, RC (resin coating).

## DISCUSSION

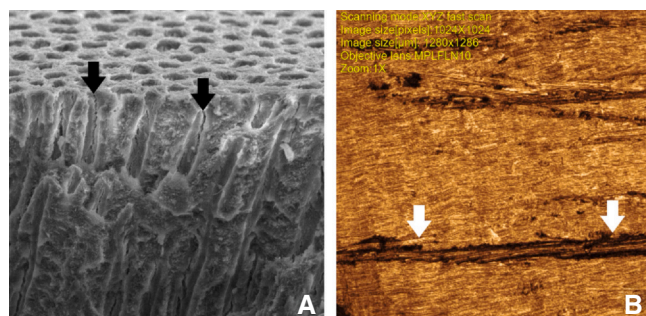
The research hypothesis that different finishing protocols would influence all tested parameters was accepted based on the significant effect of the finishing protocols on all evaluated outcomes. The research hypothesis that a reduction in dentin surface roughness and smear layer thickness would increase bond strength was rejected because of the lack of relevant correlations between

surface roughness or smear layer thickness and bond strength. The research hypothesis that resin coating would enhance bond strength to dentin was accepted, as the RC group achieved the highest bond strength values.

In the present study, all finishing protocols reduced surface roughness compared to the CT (control) group. The FB produced roughness levels similar to AO and RC. The CT and FB resulted in irregular grooves, elevated peaks, and thicker smear layer, likely associated



**Figure 4.** Laser confocal microscopy images of dentin surfaces after finishing protocols (original magnification  $\times 10$ ). A, CT (medium-grit diamond rotary instruments). B, FB (fine and superfine diamond rotary instruments). C, CB (fine and superfine tungsten carbide burs). D, UT (fine and superfine ultrasonic diamond tips). E, MS (silicon carbide and aluminum oxide with ceramic binders). F, SB (fine and superfine sintered diamond rotary instruments). G, AO (50  $\mu\text{m}$   $\text{Al}_2\text{O}_3$  particles). H, RC (resin coating).



**Figure 5.** CB finishing (fine and superfine tungsten carbide burs). A, Scanning electron micrograph (original magnification  $\times 5000$ ) showing microcracks in peritubular dentin (black arrows). B, Laser confocal micrograph (original magnification  $\times 10$ ) showing specific regions with deep valleys (white arrows).

with heterogeneously distributed diamond particles in the instruments. These factors may have negatively impacted bond strength, contributing to the high incidence of adhesive failures in dentin for the CT (54.0%) and FB (55.6%) groups.

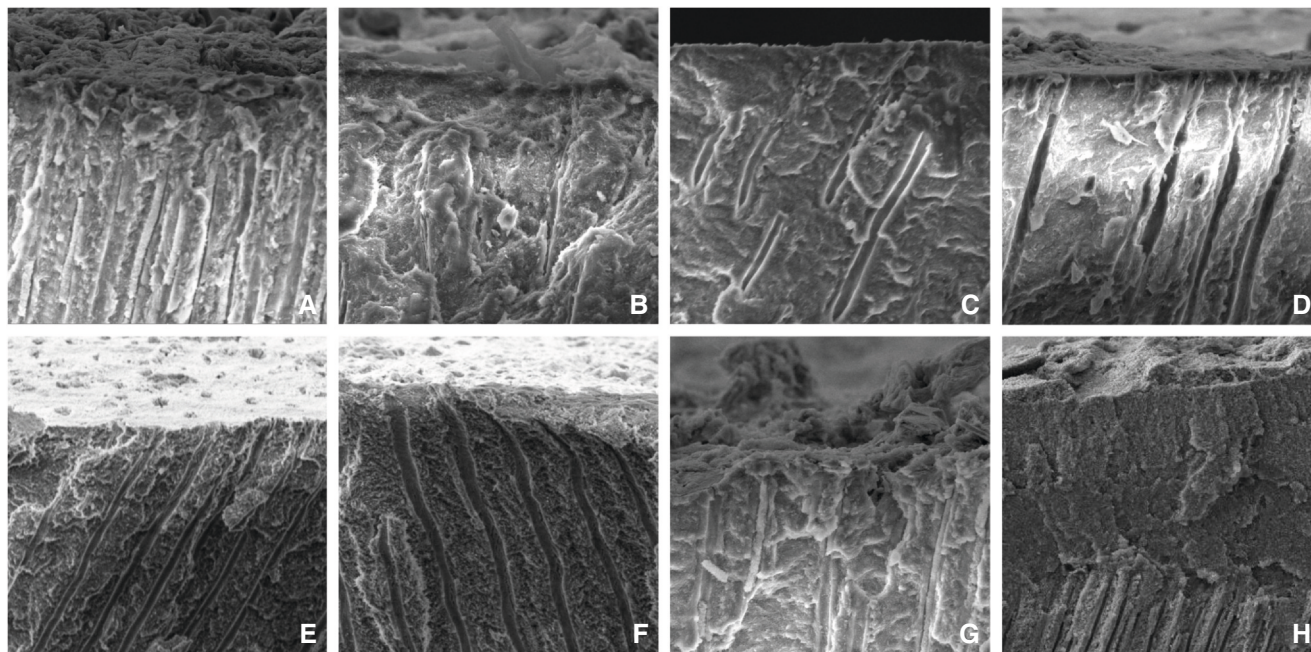
The UT resulted in lower surface roughness than FB, consistent with previous studies.<sup>1,15</sup> Beyond surface smoothing, ultrasonic finishing instruments may reduce smear layer thickness, minimize microleakage, and limit iatrogenic damage to adjacent structures.<sup>1,15</sup> However, the increased hardness of ultrasonic diamond tips fabricated via chemical vapor deposition (CVD) may

increase dentin stress, leading to microcracks.<sup>2,12,14</sup> Their rounded shape, coarse topography, and prominent diamond particles,<sup>2</sup> contribute to ultra-morphological surface damage, making the substrate more irregular and prone to adhesive failures, as seen in UT.<sup>16</sup>

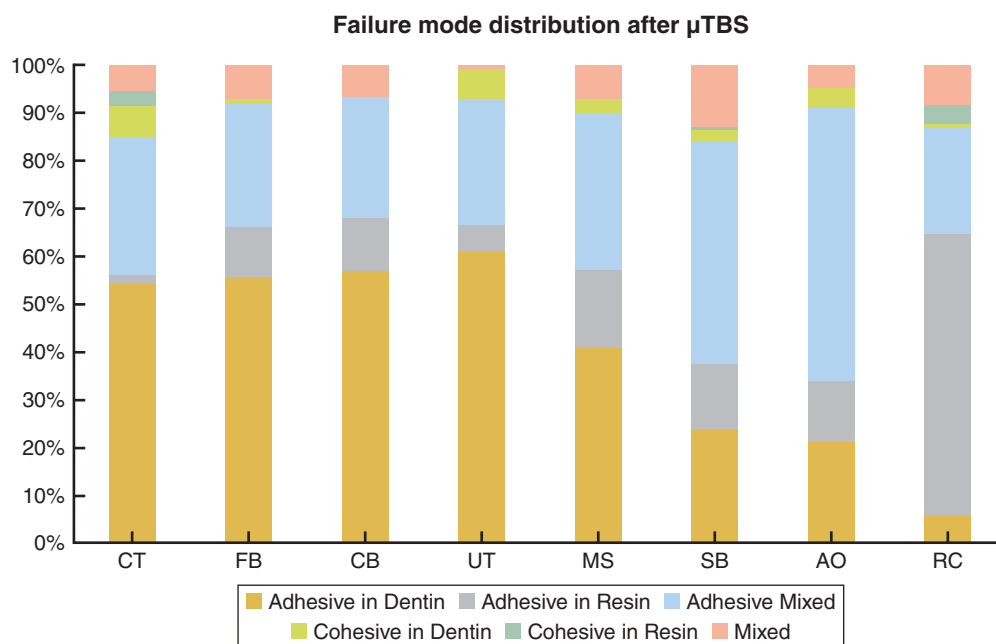
The CB resulted in lower roughness than diamond rotary instruments, likely because tungsten carbide burs cut rather than abrade hard tissues. However, a thick smear layer persisted, which may affect surface wettability and reduce bond strength.<sup>2,3</sup> The CB showed lower bond strength than most groups (except CT and SB) and a high incidence of adhesive failures in dentin (56.7%). Confocal and SEM images revealed deep valleys and microfissures in peritubular dentin (Fig. 5), likely associated with the tungsten carbide blade orientation or the 90-degree tip angle of the superfine tungsten carbide bur.

The AO produced similar bond strength and roughness to FB, consistent with previous findings.<sup>10</sup> However, AO showed higher mixed adhesive failures (57.3%), suggesting better dentin adhesion than FB, which had more adhesive failures in dentin. While airborne-particle abrasion can increase bond strength, smaller ( $\geq 30\mu\text{m}$ ) and more uniform particles at lower pressures may be more effective than the 50- $\mu\text{m}$   $\text{Al}_2\text{O}_3$  particles used in the present study.<sup>10</sup>

Mounted stones (MS), which are not well described in the literature, were tested in the present study. Composed of SiC or  $\text{Al}_2\text{O}_3$  particles embedded in



**Figure 6.** Scanning electron micrographs of smear layer formation after finishing protocols (original magnification  $\times 5000$ ). A, CT (medium-grit diamond rotary instruments). B, FB (fine and superfine diamond rotary instruments). C, CB (fine and superfine tungsten carbide burs). D, UT (fine and superfine ultrasonic diamond tips). E, MS (silicon carbide and aluminum oxide with ceramic binders). F, SB (fine and superfine sintered diamond rotary instruments). G, AO (50  $\mu\text{m}$   $\text{Al}_2\text{O}_3$  particles). H, RC (resin coating).



**Figure 7.** Failure mode distribution (%) after microtensile bond strength testing ( $\mu\text{TBS}$ ) according to dentin finishing protocols.

ceramic binders, these instruments reduced surface peaks while maintaining roughness suitable for adhesion.<sup>15</sup> Additionally, they produced a thinner smear layer than CT, FB, CB, AO, or RC, potentially explaining their higher bond strength, second only to RC. The strong integration of abrasive particles and ceramic

binders, formed at high temperatures, resulted in a continuous active tip with pores visible only at  $\times 100$  magnification.

Despite these findings, a weak correlation between surface roughness, smear layer thickness, and bond strength was observed. The SB group exhibited similar

roughness and smear layer formation to MS but had significantly lower bond strength. This could be attributed to the fact that sintered rotary instruments behaved like CVD ultrasonic diamond tips, potentially increasing dentin stress, surface microcracks, and irregularities resulting from coarser topography and greater hardness.<sup>2</sup>

The RC yielded the highest bond strength, 3 times greater than the CT (control) group. The high bond strength is particularly relevant as single-visit indirect restorations using chairside CAD-CAM workflows were simulated, without requiring surface cleaning to remove interim cement. Failures in RC occurred mostly between the resin cement and indirect restoration (59.1%), whereas in CT, FB, CB, and UT, failures were primarily adhesive to dentin, suggesting that RC effectively sealed dentinal tubules, preserving adhesion to freshly prepared dentin.<sup>17</sup> However, recent findings suggest that RC and certain finishing protocols may affect intraoral scanning accuracy because of the reflectance properties of treated surfaces.<sup>13</sup> The results indicated that FB still remain a suitable option for finishing preparations,<sup>2,3,7,8</sup> despite increased smear layer formation. However, MS or RC yielded a 50 to 75% higher bond strength than FB.

Limitations of this study included the lack of thermo-mechanical aging to assess long-term adhesive integrity and the absence of analysis on how smear layer composition may influence adhesive performance. Additionally, a self-etch, 2-step adhesive system (Clearfil SE Bond) was used, which may yield different results than total-etch or universal adhesives. In addition, only one type of resin-matrix glass-ceramic indirect restorative material was used, selected to reduce failures and simplify microtensile stick preparation compared to glass-ceramics.

Future research should explore combinations of resin coating and other finishing methods for different indirect restorations. Additionally, studies should evaluate whether RC followed by AO can mitigate light reflectance issues in intraoral scanning while maintaining strong adhesion. Further studies to assess the effects of full smear layer removal are also encouraged to better understand its role in bonding effectiveness considering dental preparations.

## CONCLUSIONS

Based on the findings of this in vitro study, the following conclusions were drawn:

1. Different finishing protocols after tooth preparation for indirect restorations influenced surface roughness, morphology, smear layer thickness, and bond strength to dentin.
2. All finishing protocols tested resulted in higher immediate bond strengths to a resin-matrix glass-

ceramic CAD-CAM material compared to the control group.

3. Resin coating protocol yielded the highest immediate bond strength to dentin across all groups.
4. Fine and superfine mounted SiC and Al<sub>2</sub>O<sub>3</sub> stones improved immediate bond strength while reducing surface roughness and smear layer thickness.
5. A very weak correlation was found between dentin surface roughness, smear layer thickness, and immediate bond strength.

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